Manuscript outline

Cell enlargement, polarity competition and migration persistence

Cell enlargement, polarity responses and heading persistence with corrected drift/jump comparisons

23 October 2026 · Figure 4: corrected event scoring and normalized drift

Scientific argument and manuscript plan

Working hypothesis: WGD-associated enlargement changes how established polarity is organized and replaced; these changes may provide a physical basis for greater migration persistence. The theoretical contribution tests size under unchanged local kinetic parameters. Length differs between model cells to represent WGD-associated enlargement, rather than elongation during a turn. Existing migration observations supply empirical motivation; new paired WGD motility results retain a substantial, explicitly reserved contribution.

Figure 2 separates size-dependent initiation and cue-width responses from temporal class signatures: initiation-limited resetting, saturating activation with visible gap retention, and pulse-2 progress after loss of visible activity. The full Holmes width series shows finite replacement at all tested lengths for sufficiently broad cues; narrow-cue censoring cannot establish a growing penalty. Cue placement relative to the incumbent’s edge switches reorientation between sliding the incumbent and takeover by a new region; enlargement slows both. Figure 3 measures time to 50% of cue-direction heading under a continuous cue and the corresponding mean angular speed; its native profiles distinguish the two regimes. Figure 4 now normalizes physical drift by the smallest size at the same input and estimates jump angles from scorable events. The existing 915,757 counted replacements include 321,423 scored angles. Event-based closure is estimable in 154 conditions; 57 are within 15% of observed decay. Among conditions with multiple fronts for most frames, 0/33 event-based and 3/33 detector-free comparisons are within 15%. The decomposition therefore retains its failures in the regime it seeks to explain. No new simulations were run. Observed migration summaries, conditional OU velocity-memory estimates and simulated polarity-heading readouts remain separate evidence.

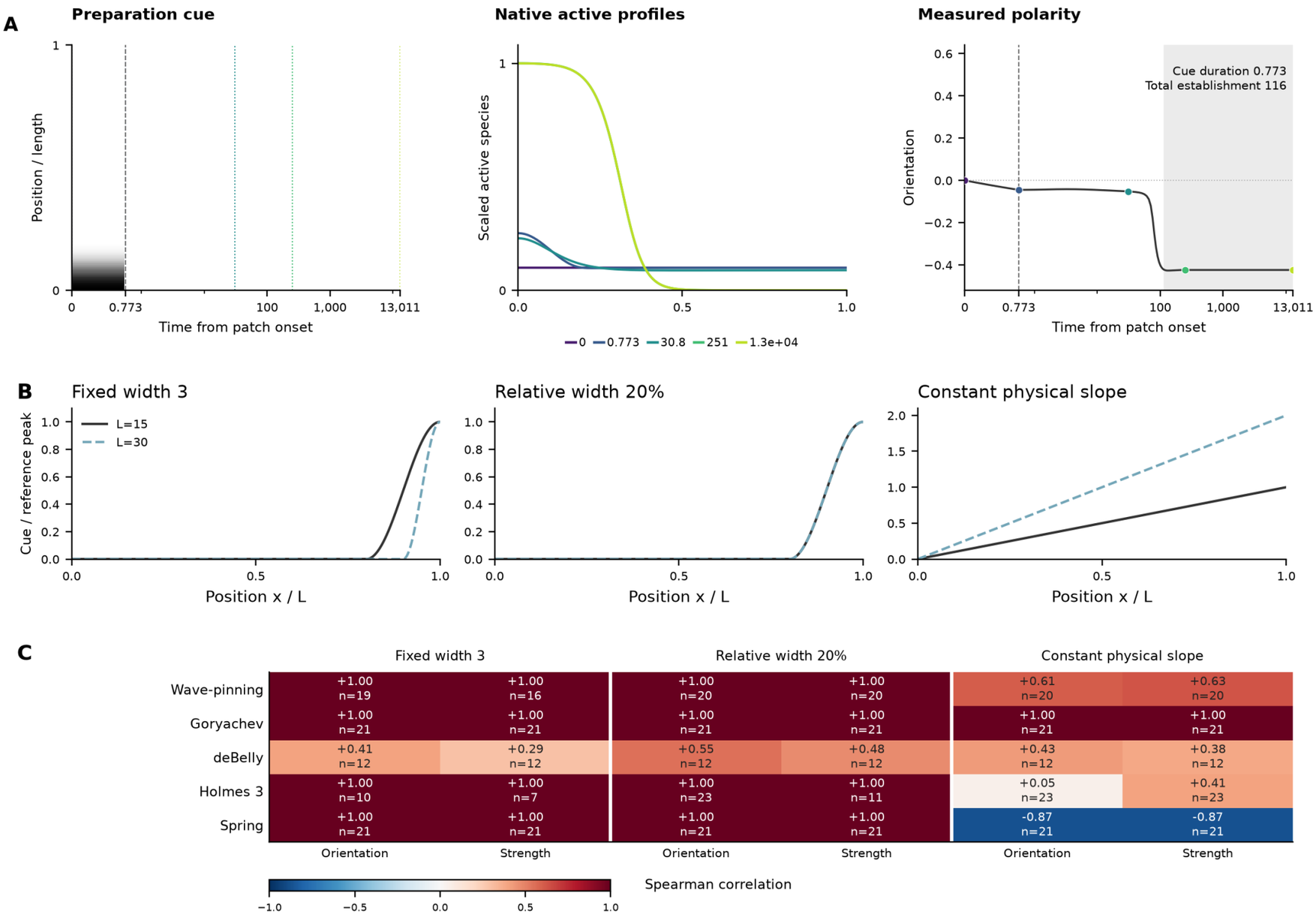
This is a working manuscript outline. Figures 1 and 5 and the reserved Figure 6 experimental section retain their preceding content. Activity polarity, modelled replacement, observed migration and conditional OU velocity memory remain distinct readouts; their biological connection is an experimental hypothesis.

Manuscript outline · continued

Proposed main figures

- Figure 1. Enlargement changes the cue duration required for lasting polarity replacement
- Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures
- Figure 3. Cue placement relative to the incumbent’s edge switches reorientation between sliding the incumbent and takeover by a new region; enlargement slows both.
- Figure 4. Null models for size-dependent heading persistence
- Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation
- Figure 6. RESERVED: new WGD motility-assay results

Figure 1. Five models and three finite challenges



Current canonical comparison, 10 October Spearman revision; corrected independent-endpoint deBelly.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement

Comparative theoretical foundation: five established models, three finite challenges, and two distinct scalar replacement gates.

Detailed legend

A, Example wave-pinning preparation from a homogeneous state using a finite left patch. The cue ends at 0.773 native time units, whereas the subsequent basal relaxation establishes the polarized profile later (reported establishment arrival about 116). Coloured profiles and orientation samples distinguish input duration from autonomous development. The grey late interval is verification, not additional stimulation. Profiles are scaled for display.

B, Three rightward finite challenges restart the identical qualifying nominal-amplitude, relative-20% basal checkpoint at each model and length: fixed physical support 3 (Spring 0.2 native length units), relative support 20%, and a linear ramp with constant physical slope. Patch peaks remain fixed within each model; a fixed-slope ramp has a larger peak in a longer domain. Goryachev patches use the independently selected, length-independent multiplier K=64 (peak 6.4), which makes this a response-permissive comparison rather than a calibrated equal biochemical dose across models.

C, Spearman rank correlation between length and the shortest tested passing pulse duration, separately for orientation replacement and strength replacement, across Wave-pinning, Goryachev, corrected deBelly, Holmes 3 and Spring. Each cell reports the correlation and included successful-length count. All audited scalar successes enter, including those failing the separate native-field tail criterion. At least four distinct successful lengths and nonconstant durations are required; censored and preparation-ineligible cases receive no invented time. The companion duration curves retain all lengths and field-status markers.

After each positive pulse, input returns to the native basal value for 10,000 native time units. Orientation requires all 201 final-500 samples to have scalar orientation above 0.05 and contrast at least 0.05, with orientation range below 0.005. Strength additionally requires every tail orientation to be within 1% of the absolute initial leftward scalar orientation; it does not compare individual pole heights or amounts. The separate native-field tail flag requires pointwise range and projected 500-unit drift within 0.1% of frozen field scales. The reported duration excludes preparation and recovery. It is the shortest passing tested duration, with sampled fail/pass boundaries refined to 1%; narrower unobserved intervals are not excluded.

deBelly uses independently commanded corrected endpoints, rather than the deprecated in-loop centered extension. Spring length and time retain native units and are not silently converted to micrometres or seconds. Model-specific axes and stimulus definitions prevent interpreting cross-model amplitudes or clocks as biologically calibrated equivalents.

Figure 1. Enlargement changes the cue duration required for lasting polarity replacement · continued

Narrative points and conclusions

- Fixed-width and relative-width patches show perfectly increasing rank order among observed successes in Wave-pinning, Goryachev, Holmes 3 and Spring; enlargement often increases the finite stimulation required for a retained response.
- The response depends on the cue geometry and model. Spring has a negative constant-slope association (Spearman correlation -0.874 for both gates), while Holmes orientation under the constant-slope cue is nearly unassociated with length (0.047).
- Corrected deBelly curves are nonmonotone despite moderate positive rank correlations. The full duration curves and eligibility ranges are essential to the result.
- These are deterministic polarity-replacement experiments. They motivate, but do not measure, spontaneous polarity memory or migration persistence.

Proposed measurements and experimental follow-up

Mechanism-linked cellular follow-up would match local input support, amplitude and duration, track the original and challenged activity regions through stimulation and withdrawal, and distinguish orientation replacement from recovered activity strength. This validation is proposed and has no experimental results in this outline.

Scope, limitations and pending validation

Correlations are conditional on observed scalar success, rather than an unbiased summary of all possible sizes or a universal scaling law. Ineligible starts are distinct from responses unobserved by the finite horizon.

A retained scalar response can coexist with a changing native field. Only 13 of the 72 corrected deBelly scalar successes pass the separate native-field tail flag. Resolution checks from the earlier centered-deBelly campaign cannot qualify the corrected replacement thresholds.

Sources and reproducibility

Three-test protocol and Spearman rule

Correlation table

Corrected endpoint method

Current duration curves

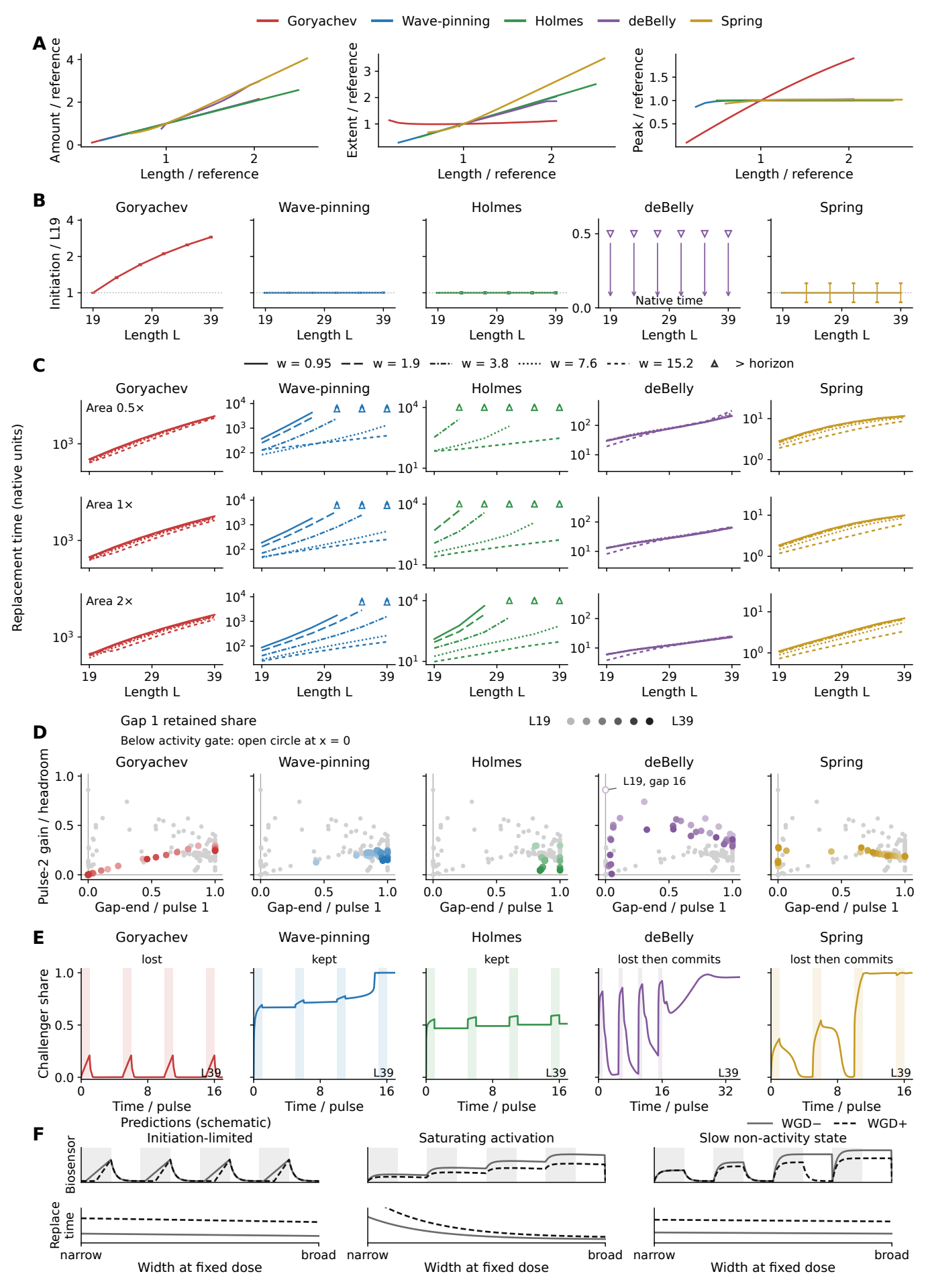

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Replacement time (native units)
Area 0.5×
Area 1×
Area 2×
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
Pulse-2 gain / headroom
Gap-end / pulse 1
L19, gap 16
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures

Separate size-dependent initiation and cue-width responses from observable pulse-memory classes.

Detailed legend

Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

A, Incumbent excess amount, extent and peak, normalized to each model's reference, across 97 retained states.

B, Initiation is the time from cue onset until maximum in-cue activity minus the profile minimum exceeds 10% of the model's L19 incumbent excess peak. Area 1, width 3.8; times are normalized to L19. All six deBelly upper bounds are 0.5 native units (downward arrows); first-update sampling leaves its size dependence unresolved.

C, Replacement time (first eligible challenger share ≥ 0.75) versus length, with a line per width and rows for areas 0.5, 1 and 2. Upward triangles/arrows are lower bounds: the cue never replaced within the horizon (6,000 Wave-pinning; 10,000 Goryachev/Holmes; 2,000 deBelly; 100 Spring, native units). Broad cues rescue Holmes across all lengths; narrow-cue censoring does not establish a growing penalty. Spring native widths equal the displayed widths divided by 38.

D, Gap-1 activity kept versus pulse-2 gain as a fraction of headroom (Eq. 1). Four equal pulses, total ON 1.5 times the model-length commitment calibration; gaps 0–16 pulse durations. Holmes L31–39 uses the L27 calibration. Other models are grey; lightness indicates L. Open circles at x = 0 code activity below the existing visibility gate. At the left wall, deBelly/Spring gain on pulse 2 despite no visible activity; Goryachev never does. The deBelly L19/gap16 first-gap endpoint was checked against native fields.

E, L39 gap4 challenger-share histories, labelled kept / lost / lost then commits; shading marks cue ON. deBelly includes full recovery; others show one post-train pulse duration. Continuous raw shares include below-gate values. deBelly's later timing and fields remain numerically sensitive (supplement).

F, Schematic biosensor predictions for the same pulse train and equal-dose width series in WGD− and enlarged WGD+ cells: delayed rises and resets; saturation with gap retention and strong width rescue; or gap collapse followed by commitment, with weaker width rescue. Predictions are prospective. Model kinetics are unchanged.

Narrative points and conclusions

- Enlarged cells resist replacement through distinct initiation, cue-width and pulse-memory signatures.

Figure 2. Replacement resistance in enlarged cells has distinct size and pulse-memory signatures · continued

- The full width series shows that Holmes can replace at every tested length when the cue is broad enough; narrow-cue censoring cannot establish a growing width-penalty ratio.
- Pulse-2 headroom gain separates low visible retention with residual progress from the initiation-limited reset response. Saturating activation retains visible activity through gaps.

Proposed measurements and experimental follow-up

Measure first in-cue biosensor rise, then vary width at fixed integrated dose.

Measure pulse-1, gap-1 and pulse-2 challenger shares under the same pulse train, followed by eventual commitment.

Scope, limitations and pending validation

Commanded input areas and native clocks are not calibrated across models.

deBelly initiation is unresolved within the first native update; later-gap timing and endpoint fields remain numerically sensitive.

D codes known below-gate activity at x=0; the exact raw ratio remains in the plotted-value CSV.

Holmes large-length trains have no target-length commitment calibration; visible retention differs from eventual replacement.

The three classes describe selected deterministic regimes; WGD biosensor predictions are prospective.

Model definitions and readouts

$$x = \frac{s_{\mathrm{gap\,1}}}{s_1}, \qquad y = \frac{s_2 - s_1}{1 - s_1} \tag{1}$$

s_1 and s_2 are challenger shares at pulse ends 1 and 2; s_gap1 is the first-gap-end share. An observed endpoint below the existing visibility gate is coded as zero visible share in D; its raw value remains in the CSV.

Sources and reproducibility

All 450 per-width replacement times

All 180 gap1/pulse2 coordinates and raw shares

deBelly L19/gap16 native-phase audit

Preserved prior main panels and complete supplement

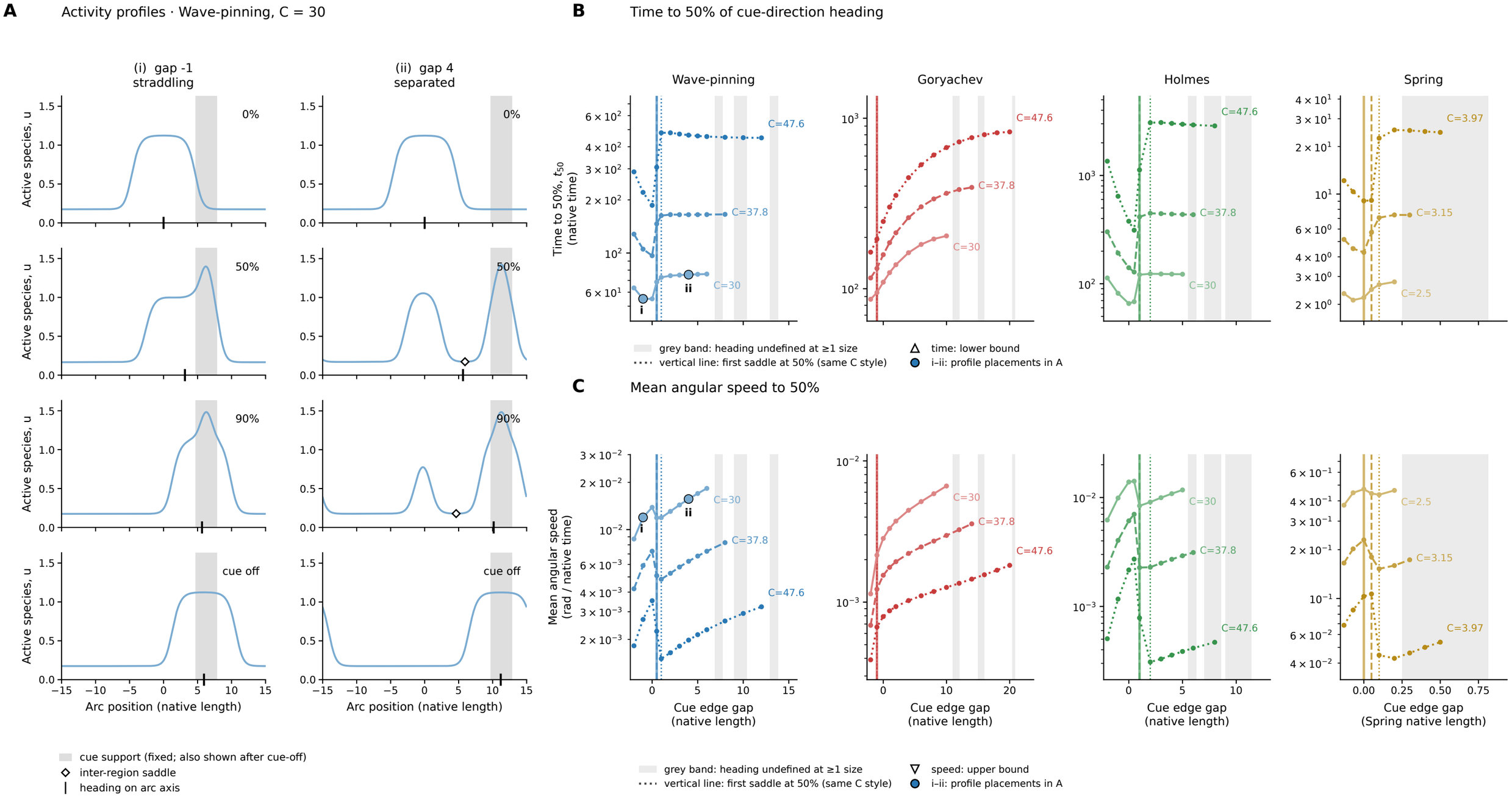

A
Activity profiles · Wave-pinning, C = 30
(i) gap -1
straddling
(ii) gap 4
separated
0%
50%
90%
cue off
Active species, u
Arc position (native length)
cue support (fixed; also shown after cue-off)
inter-region saddle
heading on arc axis
B
Time to 50% of cue-direction heading
Wave-pinning
Goryachev
Holmes
Spring
Time to 50%, t_{50}
(native time)
C=47.6
C=37.8
C=30
C=3.97
C=3.15
C=2.5
i
ii
grey band: heading undefined at ≥1 size
vertical line: first saddle at 50% (same C style)
time: lower bound
i–ii: profile placements in A
C
Mean angular speed to 50%
Mean angular speed
(rad / native time)
Cue edge gap
(native length)
Cue edge gap
(Spring native length)
speed: upper bound

Figure 3. Cue placement relative to the incumbent’s edge switches reorientation between sliding the incumbent and takeover by a new region; enlargement slows both.

Continuous-cue ring reorientation across cue placements and native sizes.

Detailed legend

A. Native Wave-pinning active species u versus arc position centred on the incumbent’s initial centre, at C = 30 for (i) gap −1 (straddling) and (ii) gap 4 (separated). Rows show 0%, 50% and 90% of the target heading under continuous cue, then the final cue-free profile. Grey shading marks the fixed cue support in every row; open diamonds mark an inter-region saddle where resolved, and black ticks on the arc axis mark heading. Each column shares its y-scale.

B. Time to 50% of the angle from initial activity-vector heading to the cue direction versus initial cue edge gap (near edge of cue support minus the incumbent’s 10%-contrast boundary). C. Mean angular speed over that turn, equal to half the target angle divided by t_{50}. Time and length use each model’s native units; speed is radians per native time. Spring retains its native mechanical units. Lighter solid, intermediate dashed and darker dotted curves denote increasing C, labelled at each line end. Enlarged filled i–ii markers identify A in both Wave-pinning plots. Both readouts are evaluated with the cue continuously on; points interpolate the saved 50% crossing brackets.

Vertical lines mark the first tested gap with two resolved regions and a saddle at the 50% crossing for each C, using that size’s line style; the existing periodic watershed detector is applied to the native saved-field interpolant. Light grey bands mark sampled near-antipode placements with undefined heading at one or more sizes (activity-vector coherence $R < 0.05$ or a cancellation interval); that size’s curve breaks there. Open up-triangles mark censored time lower bounds, and matching open down-triangles mark speed upper bounds; horizon caps are not measured values.

For the relaxation row only, cue-off occurs at 95% heading or at 20,000 if 95% is not reached, restarting from the full native event state after an identical cue-on replay. No-cue relaxation stops when max $|\partial u/\partial t|$ reaches the existing tolerance 2.2683×10^{-7} (native activity per native time; the original 10^{-4} scaled-RHS tolerance over 1,000 time units), or after four times the cue-on duration. i: cue off at 95% heading, t = 790.5; relaxed for 258.6, stopped at stationarity; ii: cue off at 95% heading, t = 227.6; relaxed for 291.3, stopped at stationarity.

deBelly is excluded from rings.

Cellular measurement motivated by this assay: photoactivation at varying offsets from the leading-edge boundary in WGD+ versus WGD− cells, with time to half-turn as the readout.

Figure 3. Cue placement relative to the incumbent’s edge switches reorientation between sliding the incumbent and takeover by a new region; enlargement slows both. · continued

Narrative points and conclusions

- At C = 30, the straddling Wave-pinning example slides the incumbent, whereas the separated example develops a second activity region and saddle. B and C retain all three sizes and show slower half-heading reorientation in enlarged rings.

Proposed measurements and experimental follow-up

Photoactivate at varying offsets from the leading-edge boundary in WGD+ versus WGD− cells; measure time to half-turn.

Scope, limitations and pending validation

Native model clocks are not calibrated across models. Topology at the 50% crossing uses saved-field interpolation; the C = 47.6, gap 0.5 Wave-pinning bracket changes topology between its endpoints. Undefined heading cannot supply a timing bound. Four fixed-parameter relaxation protocols are the only new simulations.

Sources and reproducibility

50% readouts and topology

Four native relaxation protocols

A

Drifting peak (same physical D_s) | Jumping peak (same rate) | Jumping peak (rate ∝ perimeter)

Time (×10³) — C20 — C60

Heading $C(\ell)$ — Lag (×10³)

$\alpha = 2$ (C = 20, C = 60) | $\alpha = 0$ (curves coincide) | $\alpha = -1$

B

C: 20 30 40 50 60

Drift only | Drift + same-rate jumps | Drift + perimeter-rate jumps

Local exponent α — Input fraction f

drift — same-rate jumps — perimeter-rate jumps

$f_{1/2}/f_{1/2}(C_{20})$ — Circumference C

Drift — + same-rate jumps — + perimeter-rate jumps

$\propto C$

C

Wave-pinning | Holmes | Spring | Goryachev

C: 20 30 40 50 60 | 20 30 40 50 60 | 1 1.5 2 2.5 3 | 20 30 40 50 60

C; Spring ×20 in fold panel

Local exponent α — Input fraction f

$f_{1/2}/f_{1/2}(C_{20})$ — Circumference C

Wave-pinning — Holmes — Spring — Goryachev

$\propto C$

D

Wave-pinning, $f = 0.03$ | Holmes, $f = 0.02$ | Spring, $f = 0.005$ | Goryachev, $f = 0.01$

Time (×10³) — switch — C20 — C60 — C1 — C3

Heading $C(\ell)$ — Lag (×10³)

$\alpha_{20,60} = 2.1$, 95% CI [1.9, 2.3] | $\alpha_{20,60} = 1.4$, 95% CI [1.3, 1.5] | $\alpha_{20,60} = 1.8$, 95% CI [1.7, 1.9] | $\alpha_{20,60}$ unavailable

C = 20, C = 60

E

Wave-pinning | Holmes | Spring | Goryachev

$D_s/D_s(C_{min})$

λ_{eff} — Scored events — Mixture fit

k_{pred}/k_{obs}

Multiple-front fraction

Input fraction f

Figure 4. Null models for size-dependent heading persistence

main

Detailed legend

A. Simulated drifting and relocating peaks on small and large rings. Grayscale kymograph widths follow circumference and time increases upward; each pair shares its declared seed. Ensemble heading autocorrelations below use small-ring solid and large-ring dotted curves. Fixed physical drift, same-rate relocation and perimeter-rate relocation give size exponents 2, 0 and −1, respectively.

B. Simulated mixtures illustrate how relocation erodes a geometric size advantage. Input fraction is f; the illustrative dependences are $D_s = 8.8f^2$ and $\lambda = 135f^4$, with perimeter-rate relocation multiplying λ by C/40. The decay rate is $k(\ell) = -\ln C(\ell)/\ell$; α is minus the local slope of ln k against ln circumference. Noise tolerance $f^{1/2}$ is the first downward crossing of correlation 0.5 at the fixed reference lag. Gray shades encode increasing circumference. Line styles distinguish mixtures in the tolerance-fold panel; its dashed guide is proportional to circumference.

C. Local exponents use model hues and light-to-dark circumference ramps. Markers show designed inputs, open markers identify forced conditions, vertical intervals show pointwise 95% uncertainty, and outward triangles mark off-scale values or tails. The right panel shows tolerance folds with model-specific markers; Spring circumference is multiplied by 20. 125/260 neighborhoods resolve; Goryachev contributes 11. Largest-size tolerance folds are Wave-pinning 2.373, Holmes 2.155, Spring 3.088, Goryachev 4.122.

D. Matched first-replicate model histories use the selected designed inputs, with physical widths and shared grayscale activity scales within each model. Black outside ticks identify persistent dominant-region replacements. The ensemble curves below include end-to-end exponents and 95% intervals where estimable. At every selected input, small-ring reference-lag correlation is below 0.4 and large-ring correlation exceeds 0.6; the enlarged rings retain heading longer.

E. Drift is normalized by its value at the smallest size at the same input (C = 20, or Spring 1); unity is the equal-physical-drift expectation giving α = 2. Rows below show effective jump rate, predicted/observed heading decay and multiple-front fraction. Solid curves and pointwise whole-seed intervals use counted replacement rate times mean angular loss among scorable events; dashed curves use the existing detector-free wrapped-Gaussian/uniform mixture at the shortest cadence. Unscorable events remain in the rate and exposure; their angles are represented by the scorable-event mean, with coverage and missing-angle bounds in the supplement. The jump-rate ordinate includes zero on a symmetric-log scale; open points identify forced conditions, and triangles retain off-scale ratios. Detector-free curves are point estimates. Predicted rates lie within 15% of observed rates in 57 of the 154 estimable conditions. In the multiple-front-majority regime, 0/33 event-based predictions and 3/33 detector-free predictions are within 15%. Agreement at weak input therefore does not establish closure as fronts multiply. Native

Figure 4. Null models for size-dependent heading persistence · continued

model clocks remain separate, and multiple-front fractions are descriptive.

Narrative points and conclusions

- Normalized physical drift exposes the size dependence hidden by the input-driven dynamic range. Equal physical drift produces α = 2; decreasing drift with size raises this geometric expectation.
- Predicted rates lie within 15% of observed rates in 57 of the 154 estimable conditions. In the multiple-front-majority regime, 0/33 event-based predictions and 3/33 detector-free predictions are within 15%. Agreement at weak input therefore does not establish closure as fronts multiply.
- Counted replacements and conditional angle scores now test the closure where fronts multiply and local α falls. Detector-free comparisons retain disagreement with the tracker, including positive fitted relocation terms when the tracker counts zero switches. No new simulations were run.

Proposed measurements and experimental follow-up

Measure activity-heading autocorrelation, centroid motion within persistent active regions, and persistent region replacements in WGD+ versus WGD− cells. Compare enlargement-associated changes in drift and relocation without equating activity heading to migration persistence.

Scope, limitations and pending validation

Native model clocks are not calibrated across models. Whole-seed intervals are pointwise. Missing eligible lags and tracked ingredients remain undefined; multiple-front fractions are descriptive. The required M_1/M_0 gate differs from historical round-2 normalization.

The scorable-event angle distribution estimates the angle distribution of all counted events. Coverage and missing-angle bounds are reported; this assumption is not a validation of missingness at random.

Model definitions and readouts

$$k(\ell) = -\frac{\ln C(\ell)}{\ell}, \qquad k \propto \bar{C}^{-\alpha} \tag{4.1}$$

$C(\ell)$ is activity-heading autocorrelation; $\bar{C}$ is actual mean circumference. Endpoint eligibility uses $|P| \geq 0.1$ with $P = M_1/M_0$.

$$\alpha = -\frac{\mathrm{Cov}(\ln \bar{C}, \ln k)}{\mathrm{Var}(\ln \bar{C})} \tag{4.2}$$

Figure 4. Null models for size-dependent heading persistence · continued

The local slope uses the two- or three-size neighborhood and one shared eligible lag; the lag stays fixed in whole-seed bootstrap draws.

$$k_{\mathrm{pred}} = \left(\frac{2\pi}{\bar{C}}\right)^2 D_s + \lambda\left(1 - \langle \cos \Delta\theta_{\mathrm{jump}} \rangle\right) \tag{4.3}$$

The measured drift and persistent-switch terms form a closure prediction; disagreement with observed decay is retained as a result.

$$\widehat{\lambda}_{\mathrm{eff}} = \frac{N_{\mathrm{all}}}{T_{\mathrm{dom}}} \frac{1}{N_{\mathrm{scored}}} \sum_{e \in \mathrm{scored}} (1 - \cos \Delta\theta_e) \tag{4.4}$$

All counted replacements supply the rate; scorable events supply the conditional angle distribution. The supplement reports scoring fractions and missing-angle bounds.

Sources and reproducibility

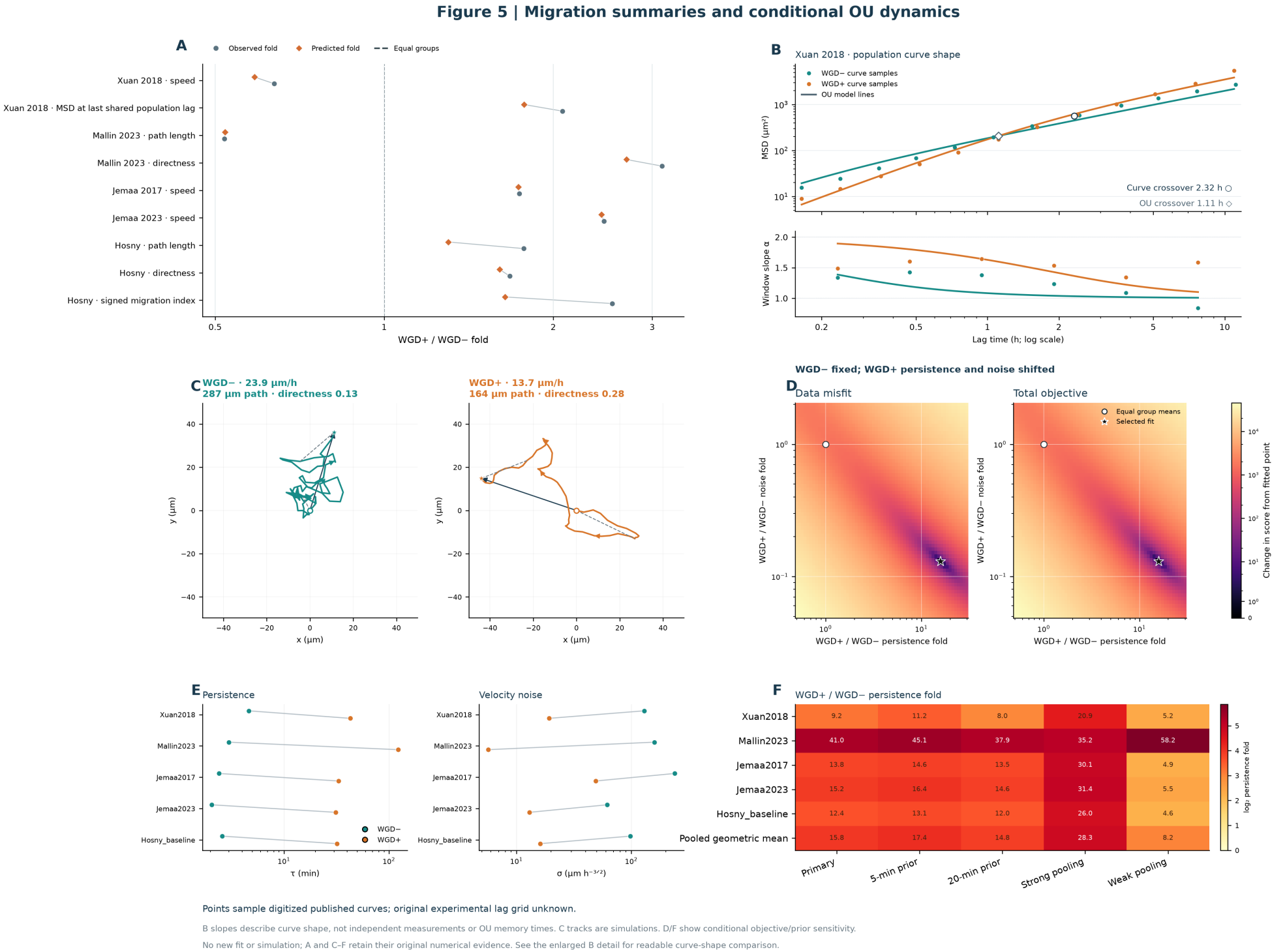
analysis receipt

null analysis receipt

campaign manifest

Event-scoring correction and regime-specific closure tests

Figure 5. Migration observations and conditional OU interpretation · image 1 of 2



Reassembled Figure 5: original A and C–F numerical evidence, revised MSD/curve-shape panel B.

Figure 5. Migration observations and conditional OU interpretation · image 2 of 2

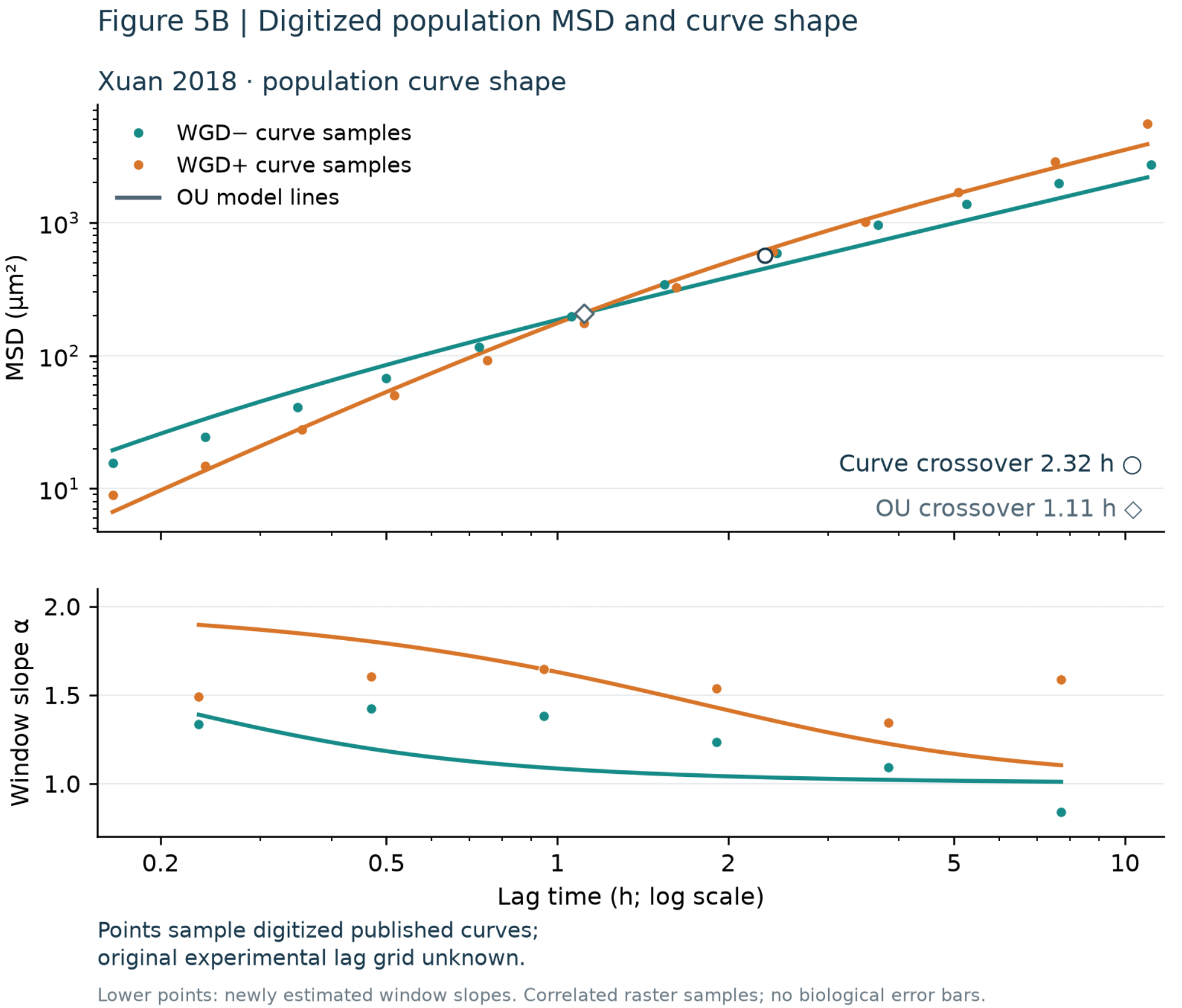


Figure 5B continued: enlarged detail with explicit raster-sample caveat.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation

Existing empirical support, with observed migration summaries kept separate from fitted velocity-memory parameters.

Detailed legend

A, Published/digitized enlarged or polyploid-group to reference-group ratios for nine migration summaries, alongside primary Ornstein–Uhlenbeck (OU) fit predictions. Xuan contributes sampled mean speed and population mean-squared displacement (MSD) at the last shared lag, approximately 10.95 h; Hosny summaries are group medians. Mallin central-bar statistic is not specified by the source and retains the historical mean approximation. These distinct source summaries are not interchangeable measures of persistence.

B, Upper: twelve chosen display samples from each digitized Xuan population MSD curve (small points) and the unchanged fitted analytical OU expectation (solid lines), on logarithmic lag and MSD axes. These are samples of the published raster curves, not recovered experimental lag positions or independent cell observations; the original experimental lag grid is unknown. The open circle identifies the crossover of the full digitized traces at approximately 2.32 h, computed by linear interpolation in native lag and MSD. The open diamond identifies the different fitted OU crossover at approximately 1.11 h. Lower: descriptive slopes of log(MSD) versus log(lag) fitted in six equal log-lag windows of the full digitized curves (points), and matching-window moving OU slopes (solid lines). Each window spans approximately 0.304 decades; both observed and model slopes use the same full window support. Observed raster columns remain correlated and give no biological sample size or confidence interval. These slopes are newly calculated curve-shape summaries, not author-reported local exponents or fitted OU memory times. A constant power-law exponent would appear horizontal; reference values 1 and 2 indicate diffusive and straight-line scaling. The model does not reproduce the complete observed curvature or the observed crossover. The analytical fit, all fitting targets and numerical inputs to A and C–F are unchanged.

C, Representative Xuan primary-fit model-generated tracks, with sampled speed, total path length and directness (net displacement divided by path length). Both groups use identical spatial scales. The tracks are simulations, not published individual-cell trajectories; a slower sampled trajectory can nevertheless have greater endpoint directness. Dashed chords illustrate the contribution of finite-time displacements.

D, With reference-group parameters fixed, vary enlarged-group persistence and velocity-noise ratios. Separate surfaces show change in the descriptive data-misfit score and in the total regularized objective; the selected fit and equal-group reference are marked. The heuristic score is not a calibrated likelihood, and the surfaces are not confidence regions.

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

E, Primary paired persistence and velocity-noise estimates for Xuan 2018, Mallin 2023, Jemaà 2017, Jemaà 2023 and the Hosny hypoxic baseline. F, Enlarged/reference persistence folds under the primary 10-minute prior, 5- and 20-minute alternatives and stronger/weaker pooling, with the pooled geometric mean. Priors and pooling define conditional sensitivity scenarios, not uncertainty intervals. The Jemaà comparisons constrain speed alone, leaving persistence/noise tradeoffs dependent on hierarchy and priors.

The fitted population contrasts span stable tetraploids, giant polyploids and treatment-induced polyaneuploid cancer cells (PACCs). The inherited WGD+/WGD− labels in the artwork are comparison shorthand and do not establish matched WGD-only interventions in every study. Hosny observations derive from the recovered preprint; equivalence to the journal version remains unresolved.

Narrative points and conclusions

- Xuan combines lower sampled speed (0.64-fold) with greater late-lag population MSD (2.08-fold); Mallin combines shorter path length (0.52-fold) with greater directness (3.12-fold). Persistence-related behaviour therefore deserves analysis separately from speed.
- Hosny baseline PACCs have greater median path length (1.77-fold) and directness (1.67-fold). Signed migration index is a separate directional summary.
- All five primary OU comparisons infer longer velocity memory. The pooled primary persistence fold is about 15.8, and remains above one across the declared prior/pooling scenarios; this is a conditional model result.
- The fit does not reproduce the complete Xuan curve shape within its raster envelope, and data-only sensitivity has rank 18 of 22. Optimization completion does not establish model adequacy or uniquely identified persistence times.
- The observed Xuan curves cross near 2.32 h while the unchanged fitted OU curves cross near 1.11 h. Matched-window slopes make the remaining curve-shape discrepancy visible; fitted persistence is a conditional summary.
- Lower sampled speed can coexist with greater late-lag displacement in the enlarged-group comparison. Its causal connection to polarity mechanisms requires matched activity and migration measurements.

Proposed measurements and experimental follow-up

The new WGD assay should measure matched trajectories directly, with declared sampling and observation duration, and assess speed, displacement, directness and lag-dependent MSD before considering fitted persistence. An activity-polarity memory

Figure 5. Published migration summaries support a persistence hypothesis with conditional OU interpretation · continued

measurement would complement these quantities without equating activity and velocity.

Scope, limitations and pending validation

Neither these heterogeneous group comparisons nor the OU fits establish that enlargement alone causes persistence. Cell identity, induction/treatment and acquisition differences require explicit treatment in the new experiments.

The requested observed-point/model-line and curve-shape display is implemented. The original experimental lag grid remains unknown; neither display samples nor raster-derived slopes supply independent biological replicates. The observed/model crossover discrepancy remains visible.

Sources and reproducibility

Refined OU analysis
Original panel legend
Observed folds
Fit results and adequacy
Hosny sensitivity supplement
Revised display, exact preserved inputs and direct-user feedback
Migration display numerical and preservation checks

Figure 6. RESERVED: new WGD motility results

RESERVED MAJOR EXPERIMENTAL RESULTS

New WGD motility assays — all data pending

A Paired biological system

SUM-159 chemically induced WGD+/− pilot

Confirm group identity, size and morphology

PENDING — no observations available

B Imaging and observed tracks

Optimize acquisition and tracking

Declare cadence, duration, eligibility and replicates

PENDING — no observations available

C Measured migration behaviour

Speed, path length, displacement and directness

Lag-dependent MSD under declared observation rules

PENDING — no observations available

D Evaluate the persistence hypothesis

Relate the acquired observations to WGD and size

Retain null or opposite outcomes; specify controls

PENDING — no observations available

Proposed panel plan only. Design and analysis remain to be agreed before interpreting data.

Explicit placeholder. Every panel is pending acquisition, analysis and author review; no mock observations are displayed.

Figure 6. RESERVED: new WGD motility-assay results

A substantial new experimental results section and main figure. No observations are available; the panel layout is a proposal, not an experimental result.

Detailed legend

A, Proposed experimental context: the SUM-159 chemically induced WGD+/− pair is the agreed starting pilot with the Speros collaboration. Imaging should first be optimized for that pair before expansion to additional cell lines. Ploidy/group identity, physical size and morphology need to be documented so the intended comparison is clear.

B, Reserved for representative experimental fields and trajectories under the selected acquisition conditions. The eventual panel should state sampling interval, usable duration, spatial scale, track eligibility and exclusions. Number of independent preparations and biological replicates remains to be specified.

C, Reserved for empirical motility summaries: sampled speed, accumulated path length, endpoint displacement/directness and lag-dependent MSD. The final readouts, follow-up window and analysis plan are to be settled before interpreting the data; this placeholder does not imply that all listed measurements will be technically feasible.

D, Reserved for the measured relationship between WGD-associated enlargement and migration behaviour, including size distributions and controls needed to interpret the group contrast. Any activity/protrusion measurements arising from the mechanism proposals require a separately agreed assay and should be distinguished from the basic motility pilot. Statistical models, uncertainty reporting and sample-size targets remain open.

Narrative points and conclusions

- This major results section will test whether the planned paired experimental system exhibits the migration behaviour motivating the theoretical work.
- Its conclusions must follow the acquired observations, including null or opposing results. Simulation output and literature fits cannot fill this gap.
- Agreement between increased polarity resistance or activity memory and migration persistence would support the proposed connection; causal attribution requires appropriate perturbations and controls.

Proposed measurements and experimental follow-up

Pending decisions include plating density, acquisition duration and cadence, field/condition throughput, tracking and exclusion rules, replicate structure, confirmation of WGD status, size quantification and the statistical analysis. The known throughput versus spatial/temporal resolution trade-off should be resolved by the pilot.

Figure 6. RESERVED: new WGD motility-assay results · continued

Mechanism validation can subsequently examine incumbent extent, early challenger growth, local loss/rebound, site fates and global activity-polarity correlation under characterized perturbations. These activity experiments are proposals alongside the theoretical mechanisms and are separate from the reserved motility-results contribution.

Scope, limitations and pending validation

No pilot acquisition, sample size, effect estimate, significance claim or experimental outcome is supplied by the available evidence.

Sources and reproducibility

Author manuscript vision and supplied collaboration context

Supplementary material and supporting analyses

Figure 4: corrected event scoring and full closure comparisons

Scorable-event coverage, missing-angle bounds, both detector-free cadences and all closure failures. Preceding editions remain linked below.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round4_20261023/figure4/figure4_supplement.pdf

Scientific methods and regime-specific counts

Figure 4: round-3 analysis and null comparisons

Pointwise and smooth size exponents, drift/jump cross-checks, sensitivity, valid fractions, replication and analytic null checks. Missing estimates and finite-rate precision shortfalls are identified in the scientific readme.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure4/figure4_supplement.pdf

Unexpected results, data status and methods

Figure 2: round-3 evidence and preserved prior panels

The complete supplement adds the per-width replacement curves and the deBelly native endpoint audit, then preserves the complete round-2 main and every earlier supplement page. Superseded fate maps, ratios and class tables are historical evidence.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure2/figure2_round3_supplement.pdf

Round-3 reader and source data

Exact preservation receipt

Figure 3: round-3 evidence and every preserved round-2 panel

The supplement retains 90/99%-heading time curves, the preceding four-placement round-3 main and legend including C = 47.6 profiles, and the complete round-2 main and supplement. Original source files remain unchanged.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_round3_20261022/figure3_single_C/figure3_round3_supplement.pdf

Round-3 reader and native source coordinates

Exact supplement preservation

Historical Figure 4: previous supplement and decision record

Historical evidence from the preceding Figure 4 edition, preserved unchanged. The comprehensive supplement retains the full designed condition grid, unavailable measurements, finite-horizon bounds, native histories, controls and the stated numerical and observational sensitivities. Earlier figure versions remain development history. S0–S12 statistical core and complete native-atlas appendix index; native pages are in all linked appendix volumes. Every numbered appendix below is part of the complete

Supplementary material and supporting analyses · continued

supplement.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/signature_figure4_20261020/supporting/vectors/figure4_revision_complete_supplement.pdf

Figure reader and source data

Independent completion review

S11 native atlas appendix 001

S11 native atlas appendix 002

S11 native atlas appendix 003

S11 native atlas appendix 004

S11 native atlas appendix 005

S11 native atlas appendix 006

S11 native atlas appendix 007

S11 native atlas appendix 008

S11 native atlas appendix 009

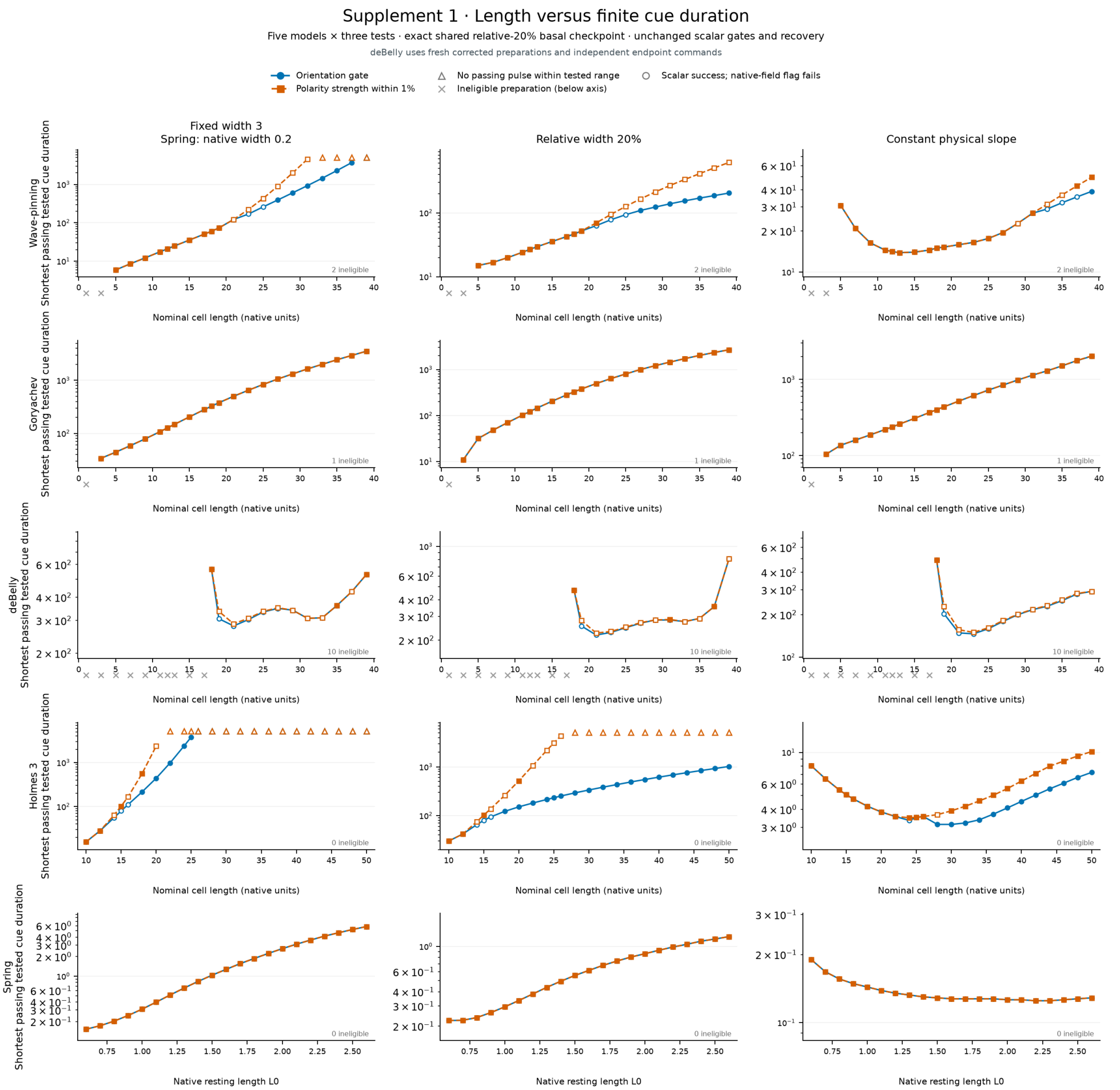
S11 native atlas appendix 010

S11 native atlas appendix 011

S11 native atlas appendix 012

S11 native atlas appendix 013

S1. Full cue-duration response curves



Both patches use a constant peak across all lengths; Goryachev K=64, peak=6.4. Gradients retain one constant physical slope per model.
Each positive duration restarts the original checkpoint. Full strength adds the 1% reversed initial-polarity band to the orientation gate.
Open markers retain failing field-tail flags. Censoring marks explicit tested limits; missing endpoints break lines. Native times and scales differ.
Data generated 2026-10-08T19:26:33.200718+00:00; producer finished. Every imported and fresh trial passed independent native-array audit.

Canonical corrected five-model duration curves; source working number Supplement 1.

Earlier supporting material: S1. Full cue-duration response curves

Essential curve context for the main comparison heatmap.

Detailed legend

Five model rows and three finite-cue columns retain the complete sampled length dependence. Coloured curves distinguish orientation from strength replacement. Open success markers retain scalar successes that fail the separate native-field tail flag; triangles indicate no passing pulse within the tested range; crosses indicate ineligible preparations. Missing values are not zero response times. Each duration excludes preparation and the subsequent 10,000-unit basal recovery. Spring uses native length and time. See Figure 1 for exact gates and stimulus conventions.

Narrative points and conclusions

- Corrected deBelly is nonmonotone across all three geometries. The positive rank coefficients summarize association rather than its full curve shape.
- Constant-slope Spring shows the opposite association to its patch responses. Holmes orientation under that cue varies with little monotone association. Censoring and native-field flags restrict interpretation even where rank correlation is strong.

Sources and reproducibility

Current figure legend and gate definitions

Supplementary material and supporting analyses

Earlier supporting material: S2. Numerical qualification and preparation provenance

Exact common starting states, duration-boundary audits, corrected deBelly endpoint controls and selected mesh diagnostics; source histories remain distinguishable from compact committed summaries.

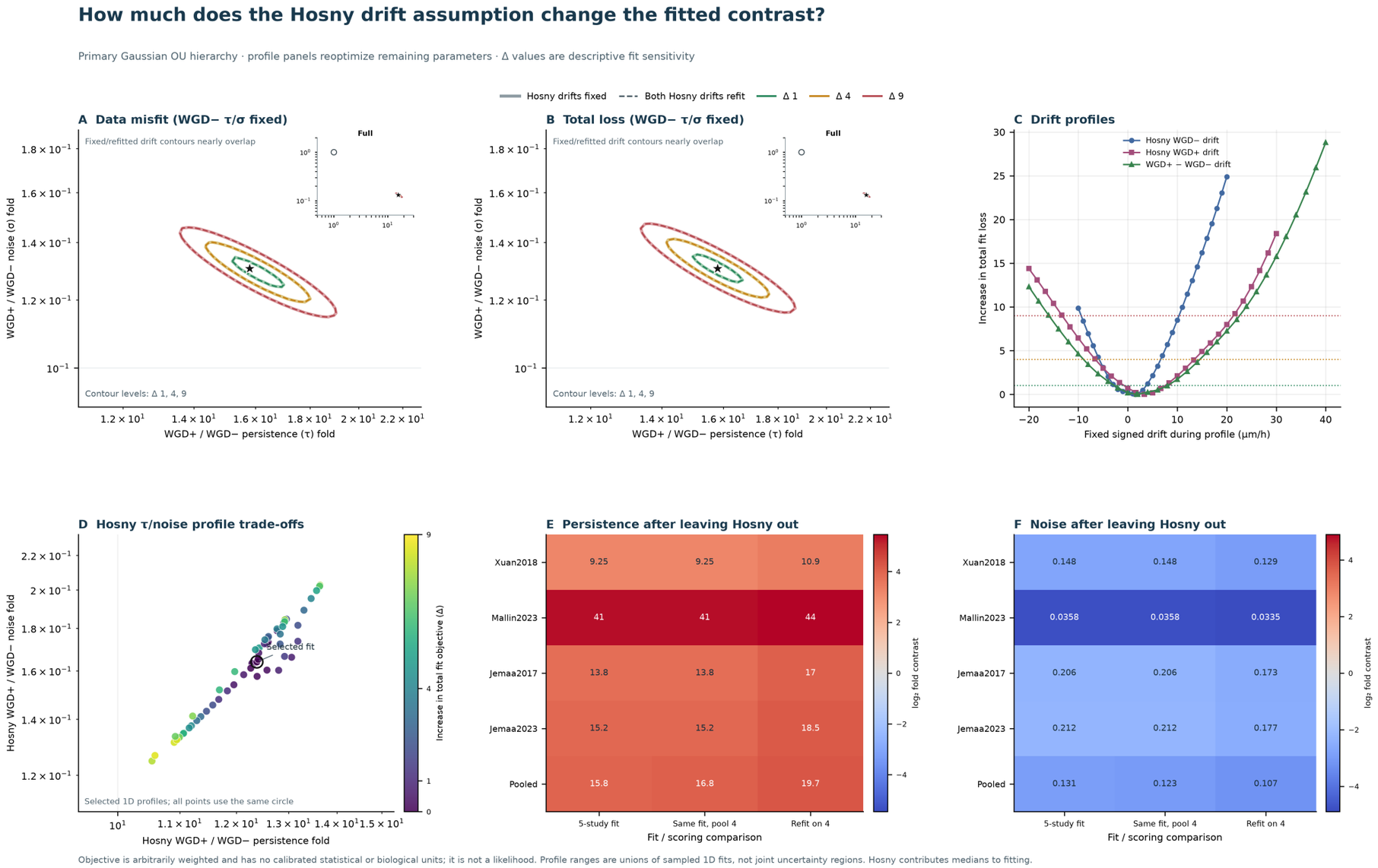
polarity1d/manuscript/THREE_TESTS.md

Earlier supporting material: S3. Mechanism and stochastic sensitivity

Complete gain/recovery grids, clock qualification, finite-cue definitions, outcome tables, site/vector diagnostics and numerical/noise controls accompany the mechanism checkpoint. The historical Figure 4 edition also displays both strengths, complete half-window diagnostics, physical-site eligibility and matched C20/C60 native examples; compact source tables support portable rerendering.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/VISION_CHECKPOINT_REQUIREMENTS.md

S4. Migration-model adequacy and Hosny sensitivity



Existing Hosny sensitivity supplement from the refined OU analysis.

Earlier supporting material: S4. Migration-model adequacy and Hosny sensitivity

Supporting conditional sensitivity analysis; it is not an independent biological replication.

Detailed legend

The retained supplement explores signed-drift and leave-Hosny-out sensitivity within the declared OU hierarchy. Its fitted changes and descriptive-score profiles are conditional on the model and supplied summaries, not confidence intervals. Data-only identifiability, Xuan population-curve residuals and prior/pooling assumptions remain part of the adequacy assessment. Representative movie diagnostics are separate: acquisition timing is uncertain and nuclear appearance does not establish ploidy.

Sources and reproducibility

Original Hosny supplement caption

Sensitivity methods and results

Earlier supporting material: Gain and small-loss recovery: supporting evidence

Detailed legend

Calculated native input gain is separate from observed post-cue activity. The small-loss recovery contrast persists through stored finite trajectories but weakens with time; actual damage is only about0.22–0.285% of core activity, with no reporter calibration.

Scope, limitations and pending validation

Small-loss assays are not the finite optical challenge or a universal stability measure.

Sources and reproducibility

Finite-time and amplitude audit

All97 gain/recovery values

Earlier supporting material: Selected native pulse and causal controls

Detailed legend

WP L15 T20/T60 starts from the same basal state and illustrates withdrawal-dependent domain survival. Evolved interventions have endpoint- and cue-specific limits.

Scope, limitations and pending validation

WP4× mobility passes orientation only at one dose; both conditions fail the stronger full-polarity-strength gate. Mobility is changed during cue and recovery.

Sources and reproducibility

Retained WP short/long pulse artwork

Endpoint-qualified controls

Supplementary material and supporting analyses

Earlier supporting material: Stationary-cue endpoint-vector alternative

An alternative view displays all 240 measured endpoint vectors on equal coordinate scales, with weak-resultant, overlap and finite-tail flags retained. It is available for discussion without replacing the complete withdrawal/retention map in the current review.

polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure3/standard_angles_sizes/vector_grid/README.md

Earlier supporting material: Spring: every native medoid

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_all_histories.png

Earlier supporting material: Spring magnitude, geometry and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_magnitude_geometry_controls.png

Earlier supporting material: Spring full/half-window memory

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_memory_windows.png

Earlier supporting material: Spring increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_increment_tails.png

Earlier supporting material: Spring short/long-lag geometry and circular mixing

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_endpoint_lag_sensitivity.png

Earlier supporting material: Spring largest sampled increment native context

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_large_increment_context.png

Earlier supporting material: All chemical models including Goryachev

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_stochastic_revision.png

Earlier supporting material: Chemical memory: both strengths

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_both_strengths_memory.png

Supplementary material and supporting analyses · continued

Earlier supporting material: Chemical magnitude and controls

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_magnitude_controls.png

Earlier supporting material: Chemical increment tails

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_increment_tails.png

Earlier supporting material: Chemical site-gate sensitivity

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_site_gate_sensitivity.png

Earlier supporting material: Original chemical medoids, all18 conditions

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_selected_histories.png

Earlier supporting material: Goryachev lowest-resultant diagnostic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/vision_noise_20261017/vision_noise_lowest_resultant_diagnostic.png

Earlier supporting material: Spring active amount and spatial first harmonic

/home/4473331/projects/scaling_sandbox/polarity1d/manuscript/figures/length_mechanisms_20261008/dev/revision_program_20261019/figure4/figure4_spring_amount_spatial_balance.png

Earlier supporting material: Preserved complete mechanism and assay checkpoint

The preceding outline retains all-five mechanism-to-measurement sheets, ring gain/recovery and finite-cue maps, and noise magnitude, controls and full/first/last-window diagnostics. Its numerical evidence is preserved; the revised main-figure claims are stated in this edition.

manuscript_outline/outline_v2.pdf

Prior all-five mechanism evidence crosswalk

Prior complete ring and noise checkpoint

Prior physical motion, magnitude and coverage evidence

References and evidence provenance

Reviewed primary literature and extraction map
Scientific scope and author direction
Direct-user preference evidence and foundational figure audit
Mechanism checkpoint intake and live-producer evidence
Source checkout, evidence selection and reproduction
Independent critical assessment and repair decisions
Portable original/renumbered mechanism replay and source closure
Second-edition integration decisions and validation scope
Figures 2–4 revision specification
Figure 2 October 20 revision brief
Figure 3 October 20 revision brief
Figure 4 October 20 revision brief